

Programming Assignment 1: Introduction to Computer Networks 512.4662

Students:

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LLM Attribution

Parts of this code were developed with the assistance of an LLM (Kiro/Claude). All generated code was reviewed, tested, and understood by the authors.

Submission Contents

- `server.c` : Implementation of the dual-thread UDP server using POSIX threads and `sys/queue.h` for dropping-tail queue management.
- `client.c` : Implementation of the UDP client that generates job demands via Poisson process distributions.
- `Makefile` : Compilation script configured with strict safety and sanitization flags.
- `README.pdf` : This document containing experiment statistics, graphics, and system analysis.

Experiment Results

1. Single Client, Unbounded Queue

(Queue size set to >4000 to guarantee unbounded behavior)

Parameters: ($\mu=5$, $\lambda=3$, Jobs=1000)

- **Average Job Time:** 505902.00 ns
- **Median Job Time:** 352000.00 ns
- **Average Queue Occupancy:** 1.53
- **Median Queue Occupancy:** 1.00

Parameters: ($\mu=3$, $\lambda=5$, Jobs=1000)

- **Average Job Time:** 63550044.00 ns
- **Median Job Time:** 68117500.00 ns
- **Average Queue Occupancy:** 157.56
- **Median Queue Occupancy:** 176.00

Parameters: ($\mu=50$, $\lambda=30$, Jobs=1000)

- **Average Job Time:** 49264.00 ns
- **Median Job Time:** 36000.00 ns
- **Average Queue Occupancy:** 1.40
- **Median Queue Occupancy:** 1.00

Parameters: ($\mu=5$, $\lambda=3$, Jobs=4000)

- **Average Job Time:** 544094.75 ns
- **Median Job Time:** 384000.00 ns
- **Average Queue Occupancy:** 1.67
- **Median Queue Occupancy:** 1.00

Parameters: ($\mu=3$, $\lambda=5$, Jobs=4000)

- **Average Job Time:** 266744677.25 ns
- **Median Job Time:** 255642000.00 ns
- **Average Queue Occupancy:** 651.09
- **Median Queue Occupancy:** 637.00

Parameters: ($\mu=50$, $\lambda=30$, Jobs=4000)

- **Average Job Time:** 52041.50 ns
- **Median Job Time:** 38000.00 ns
- **Average Queue Occupancy:** 1.46
- **Median Queue Occupancy:** 1.00

Parameters: ($\mu=50$, $\lambda=35$, Jobs=2000)

- **Average Job Time:** 57291.00 ns
- **Median Job Time:** 41000.00 ns
- **Average Queue Occupancy:** 1.63

- **Median Queue Occupancy:** 1.00

Parameters: ($\mu=50$, $\lambda=40$, Jobs=2000)

- **Average Job Time:** 65153.00 ns
- **Median Job Time:** 47000.00 ns
- **Average Queue Occupancy:** 1.81
- **Median Queue Occupancy:** 1.00

Parameters: ($\mu=50$, $\lambda=45$, Jobs=2000)

- **Average Job Time:** 83275.50 ns
- **Median Job Time:** 55000.00 ns
- **Average Queue Occupancy:** 2.45
- **Median Queue Occupancy:** 1.00

2. Two Clients, Unbounded Queue

(2000 jobs each, $\mu=50$, $\lambda=20$)

- **Average Job Time:** 84508.75 ns
- **Median Job Time:** 62000.00 ns
- **Average Queue Occupancy:** 2.57
- **Median Queue Occupancy:** 2.00

3. Single Client, Bounded Queue (Size = 10)

(2000 jobs, checking dropping behavior)

Parameters: ($\mu=50$, $\lambda=45$)

- **Total Jobs Dropped:** 15
- **Percentage Dropped:** 0.75%
- **Average Job Time:** 80804.53 ns

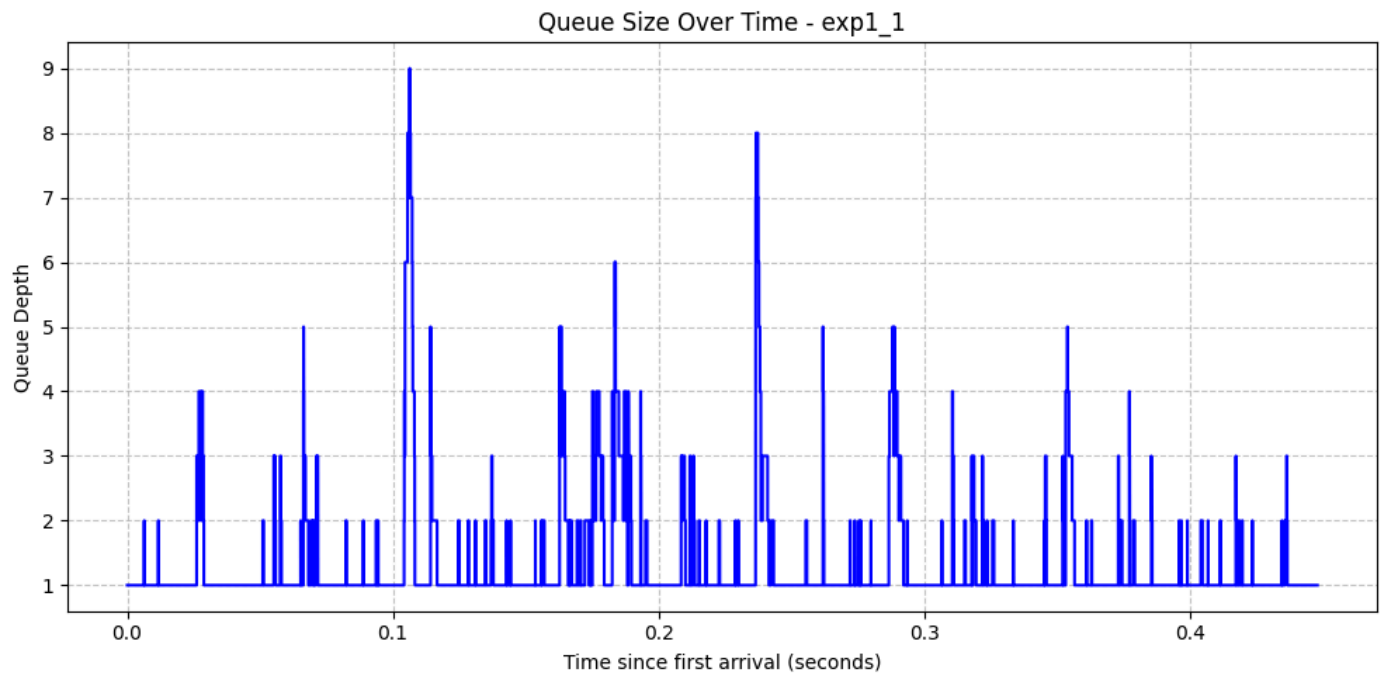
Parameters: ($\mu=50$, $\lambda=48$)

- **Total Jobs Dropped:** 28
- **Percentage Dropped:** 1.40%
- **Average Job Time:** 88773.83 ns

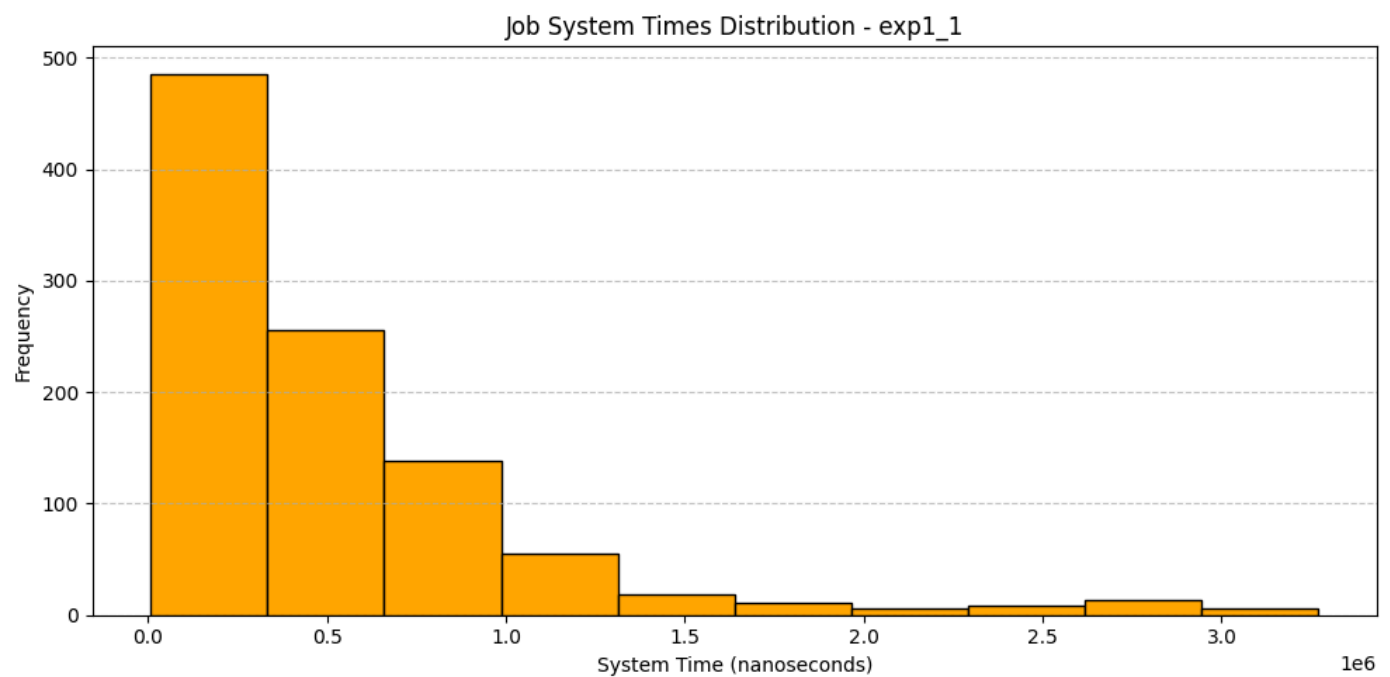
Visualizations

1. Single Client, Unbounded Queue ($\mu=5$, $\lambda=3$, Jobs=1000)

1. Queue Size Over Time

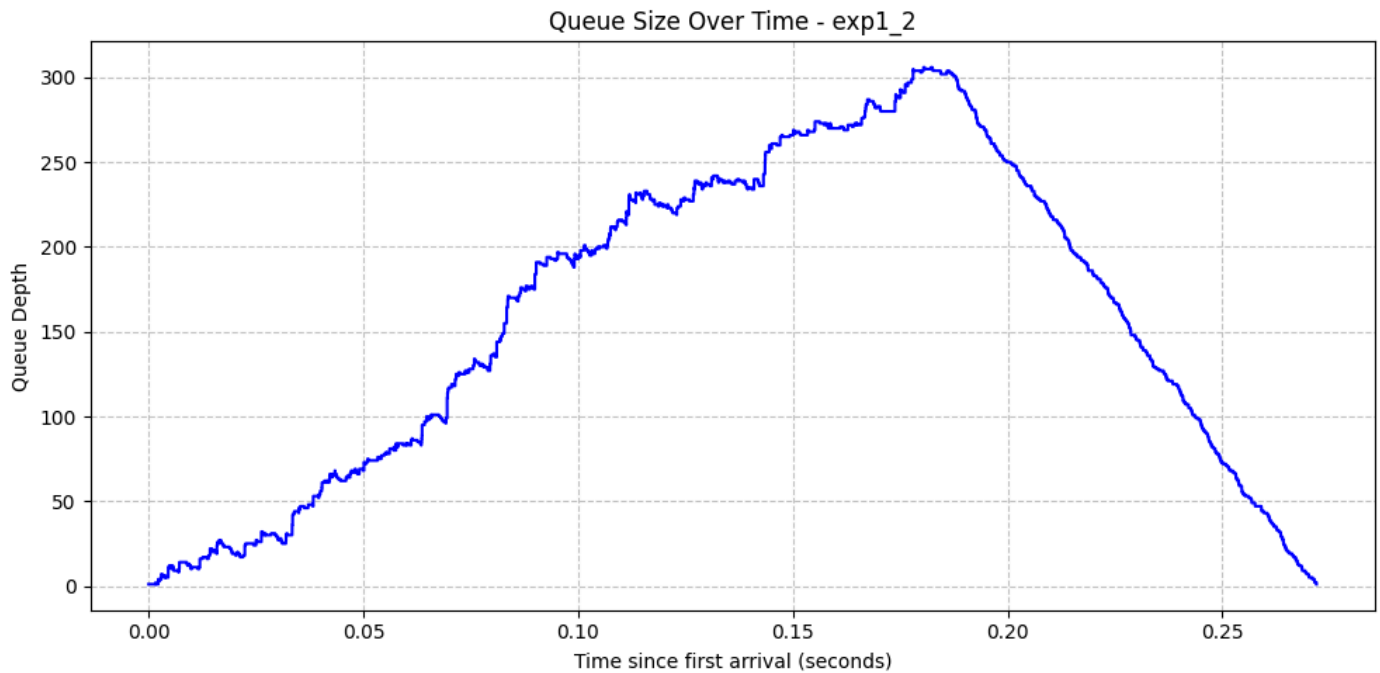


1. Job System Times Histogram

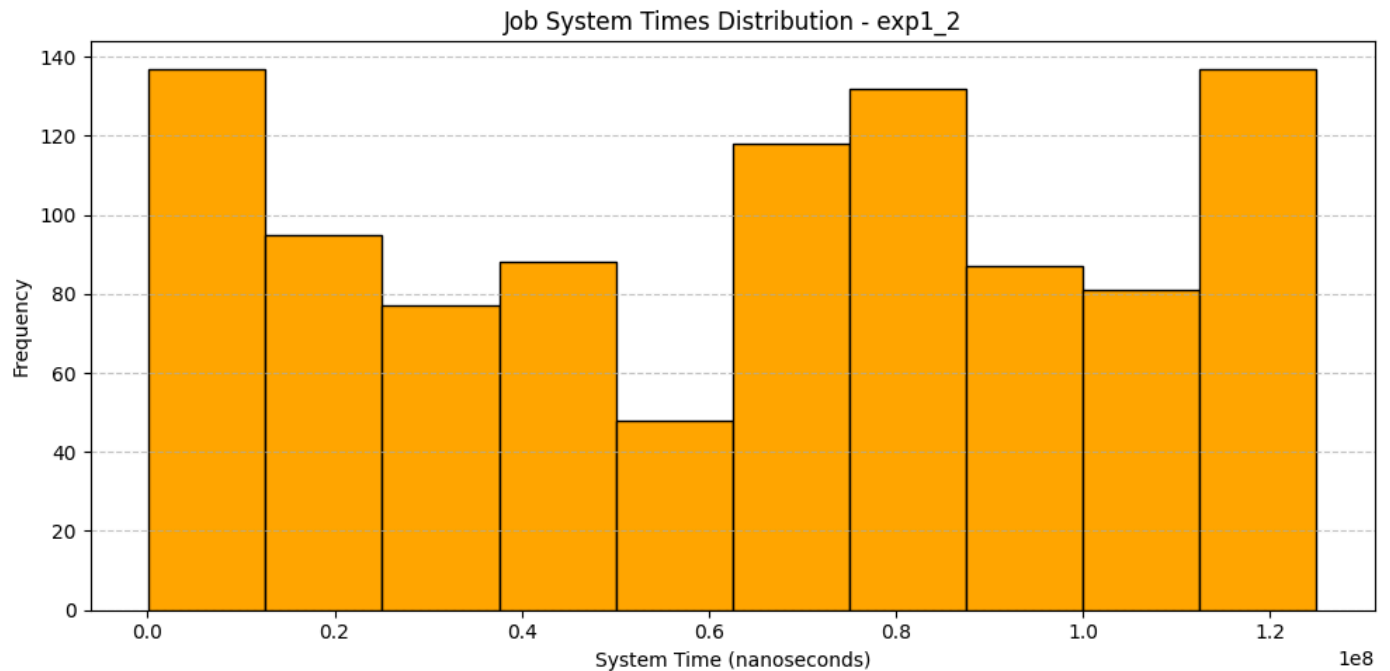


2. Single Client, Unstable Unbounded Queue ($\mu=3$, $\lambda=5$, Jobs=1000)

2. Queue Size Over Time

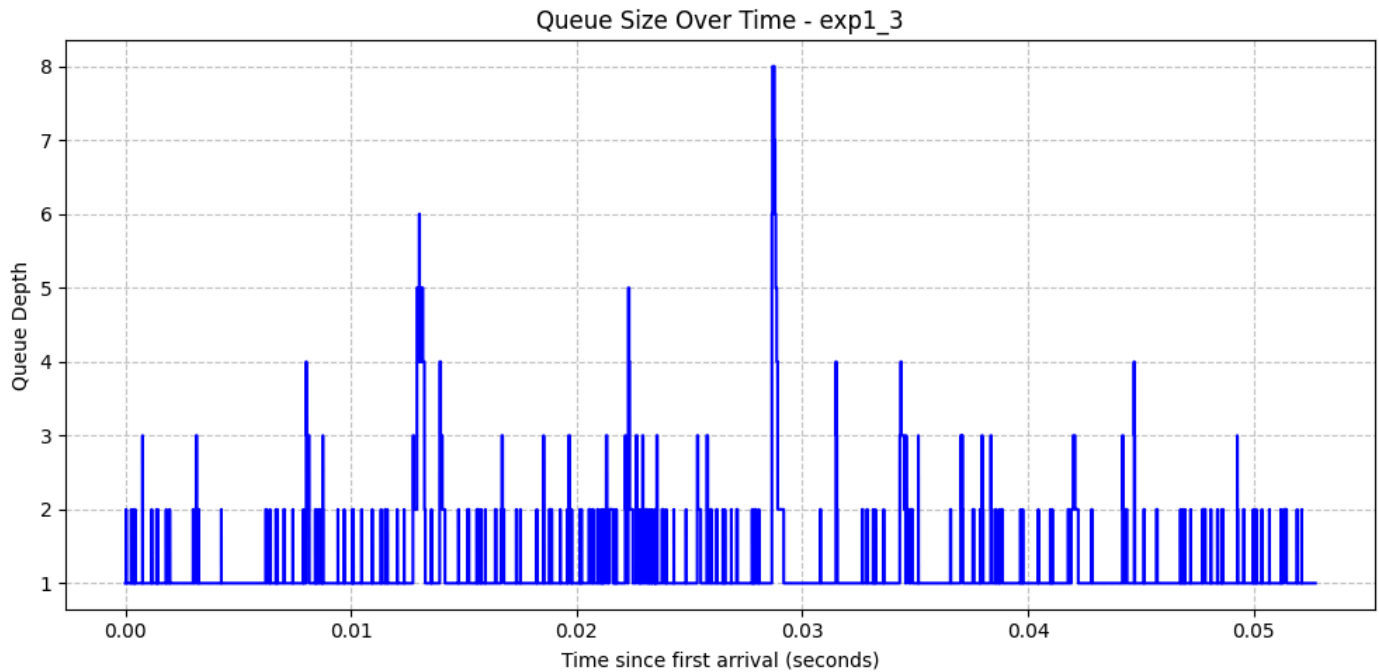


2. Job System Times Histogram

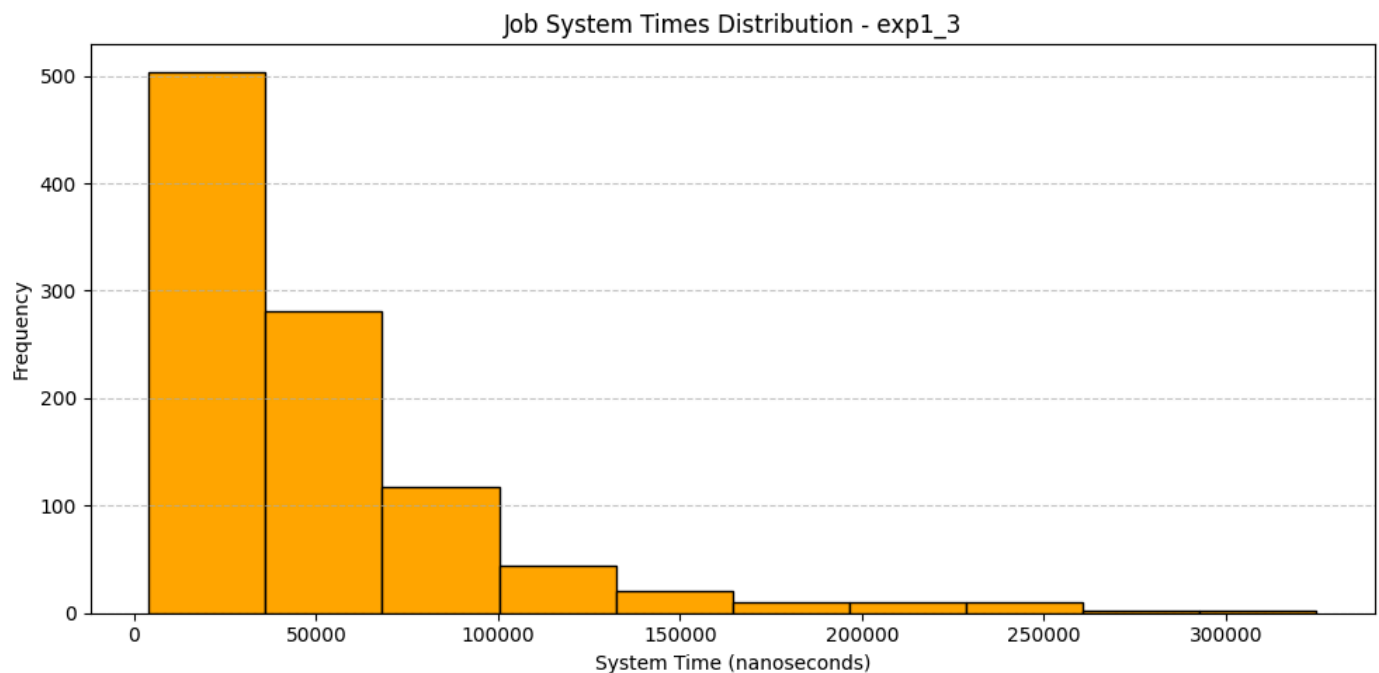


3. Single Client, Unbounded Queue ($\mu=50$, $\lambda=30$, Jobs=1000)

3. Queue Size Over Time

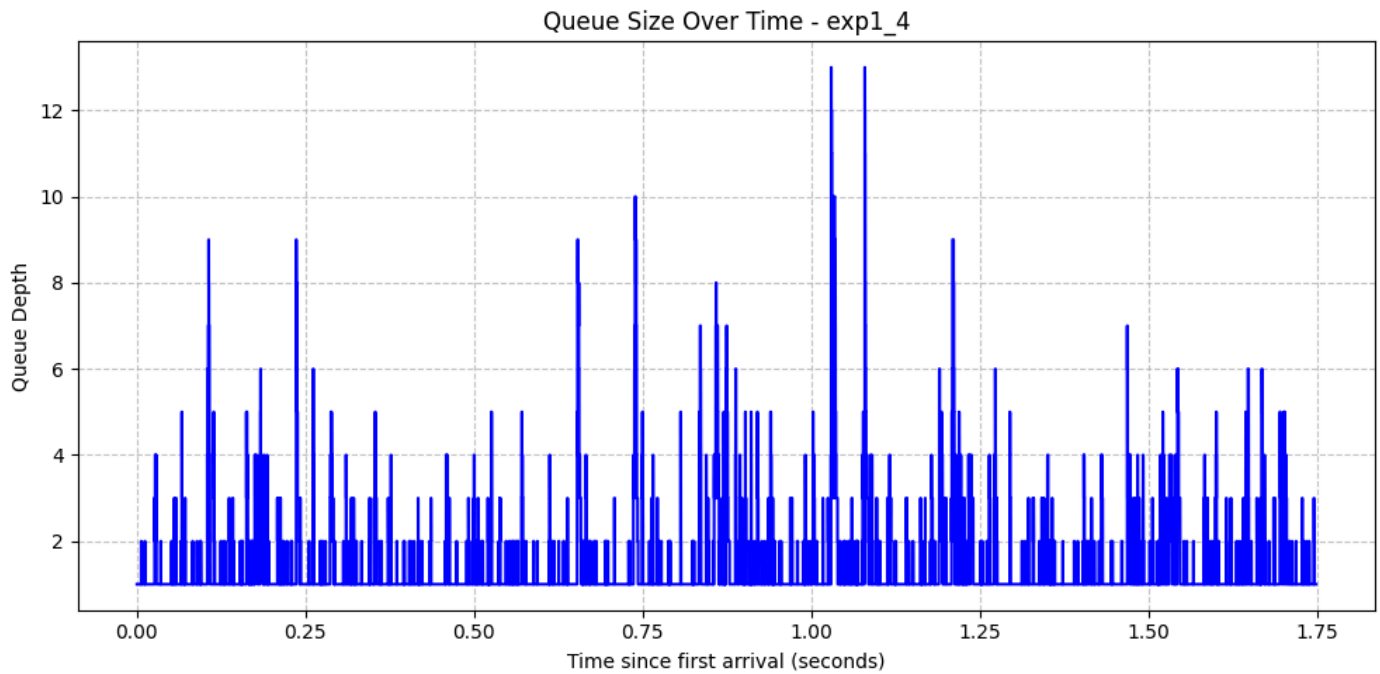


3. Job System Times Histogram

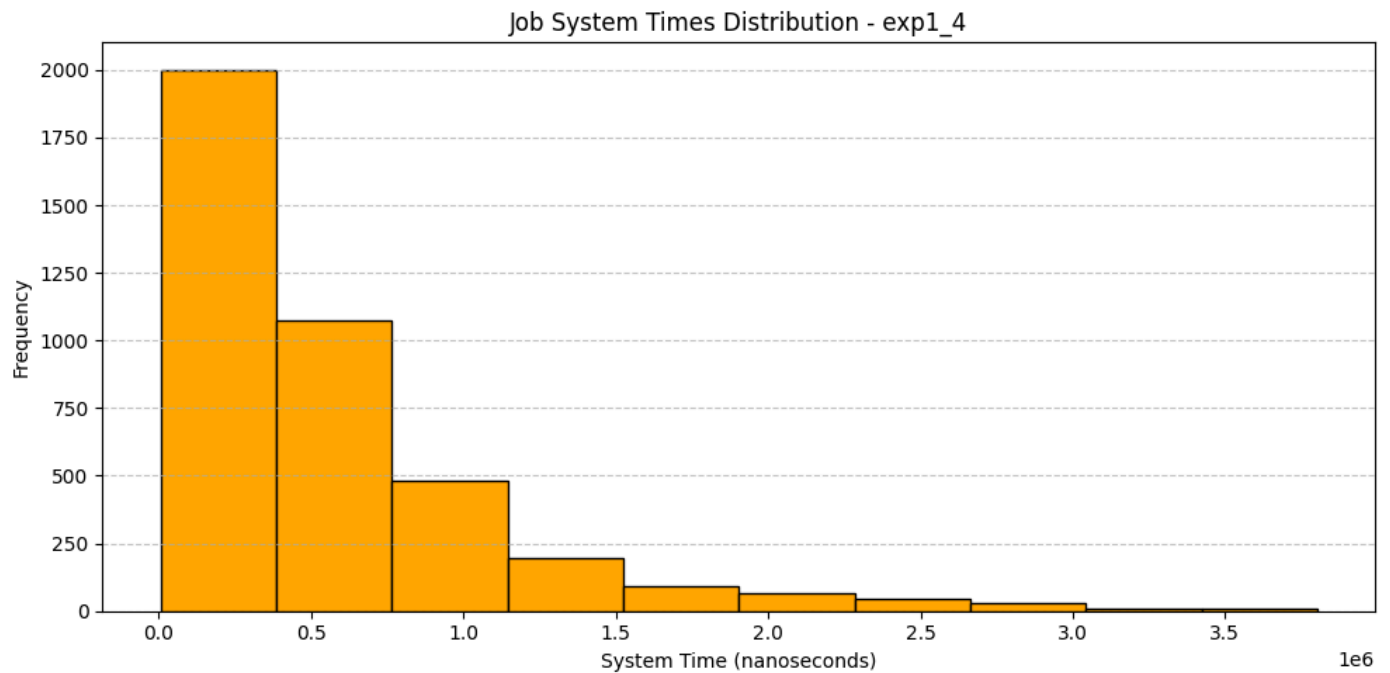


4. Single Client, Unbounded Queue ($\mu=5$, $\lambda=3$, Jobs=4000)

4. Queue Size Over Time

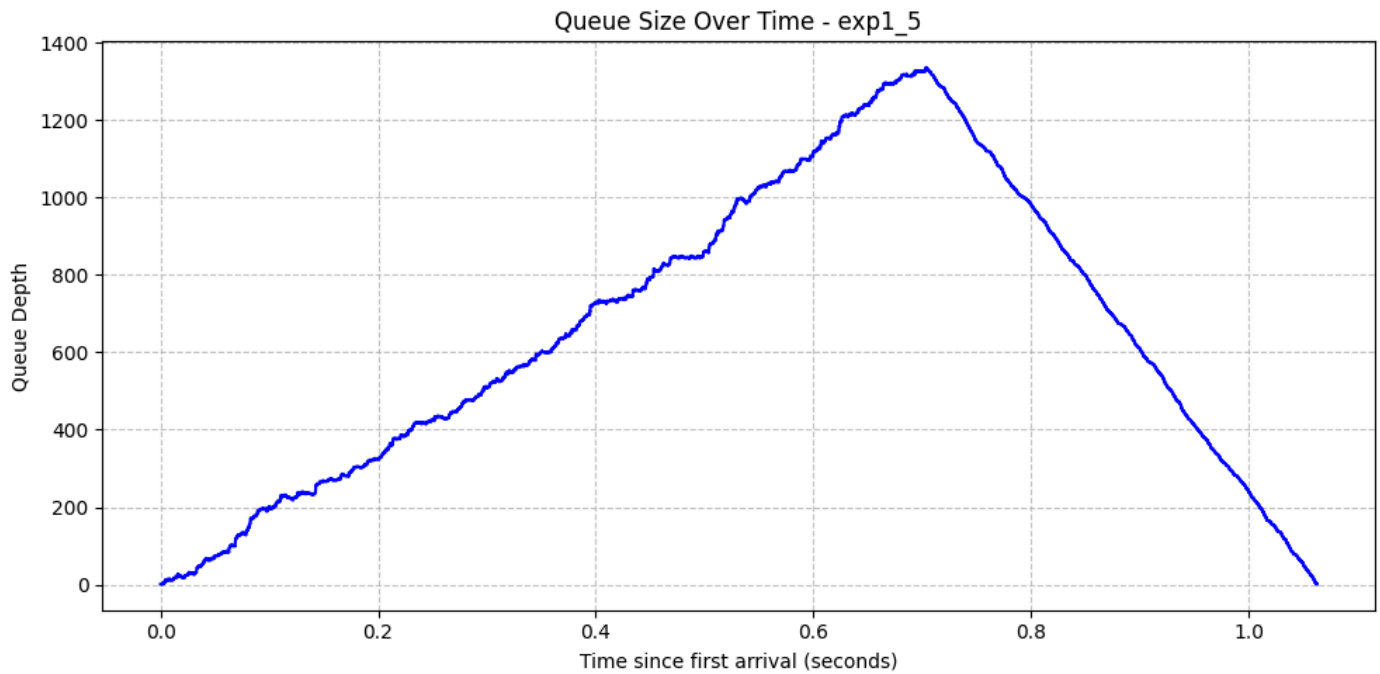


4. Job System Times Histogram

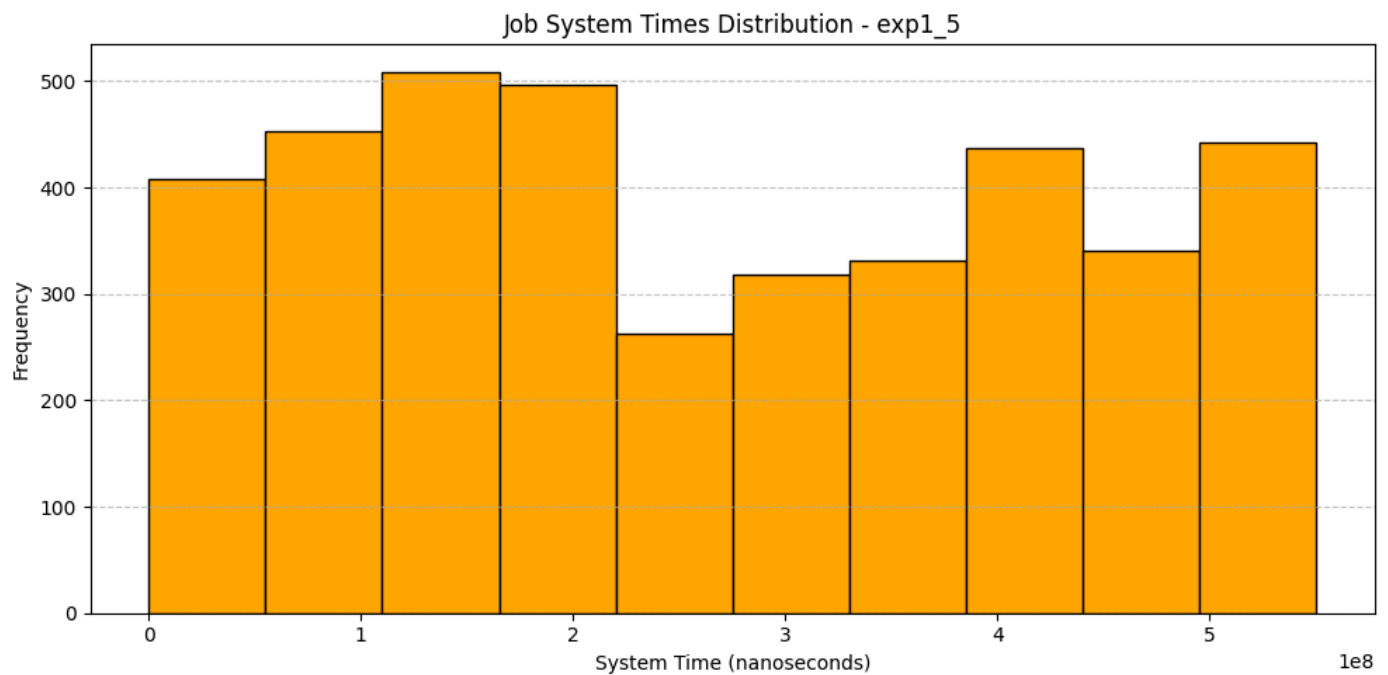


5. Single Client, Unstable Unbounded Queue ($\mu=3$, $\lambda=5$, Jobs=4000)

5. Queue Size Over Time

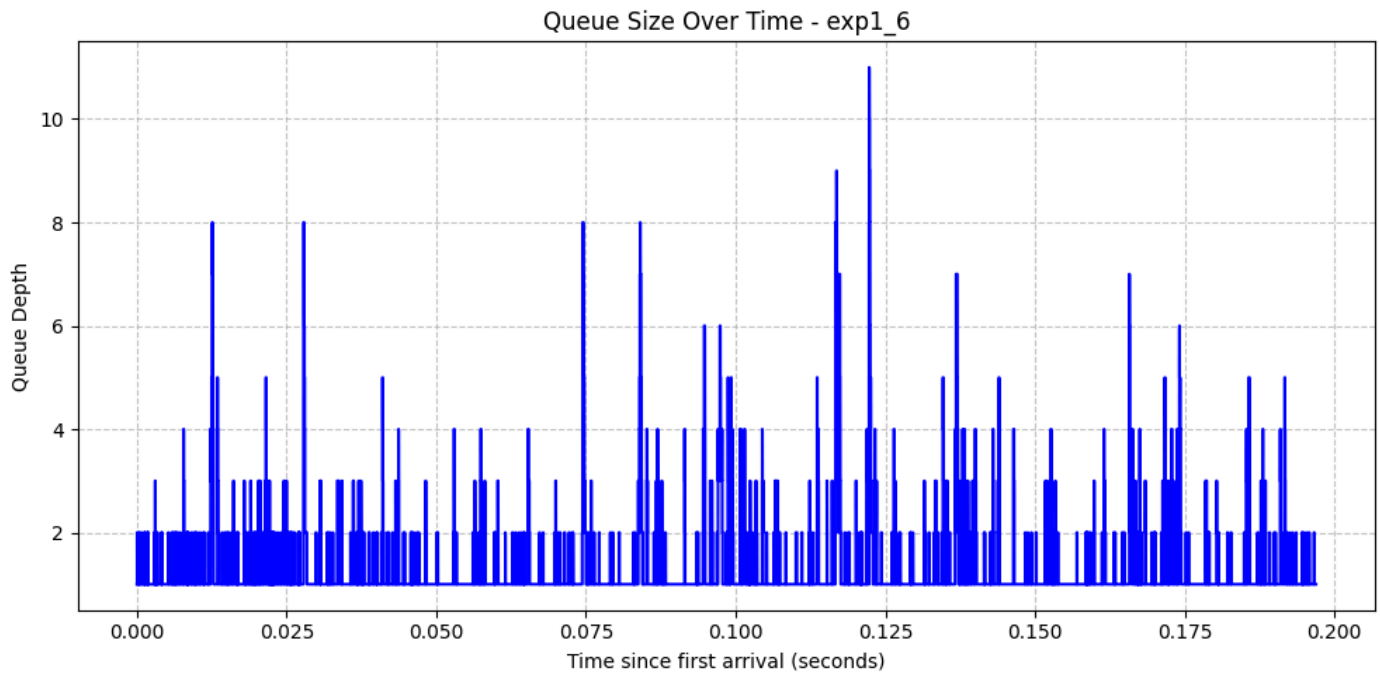


5. Job System Times Histogram

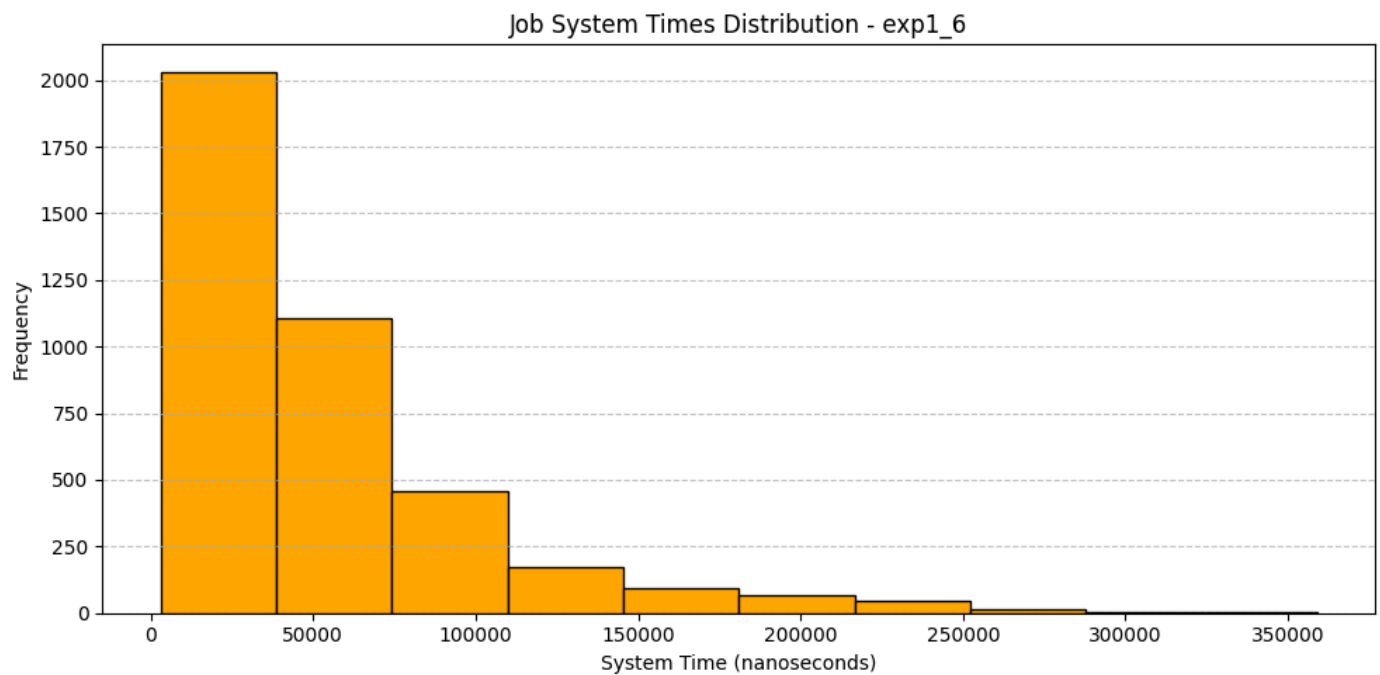


6. Single Client, Unbounded Queue ($\mu=50$, $\lambda=30$, Jobs=4000)

6. Queue Size Over Time

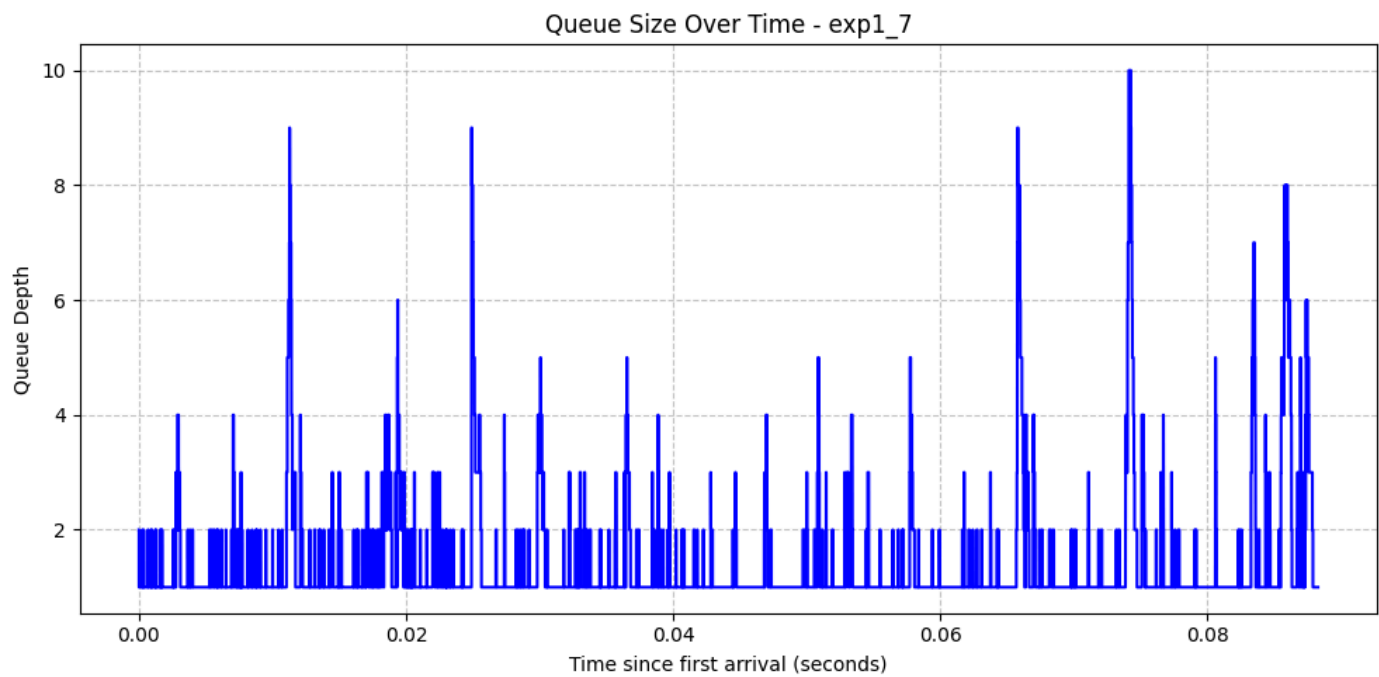


6. Job System Times Histogram

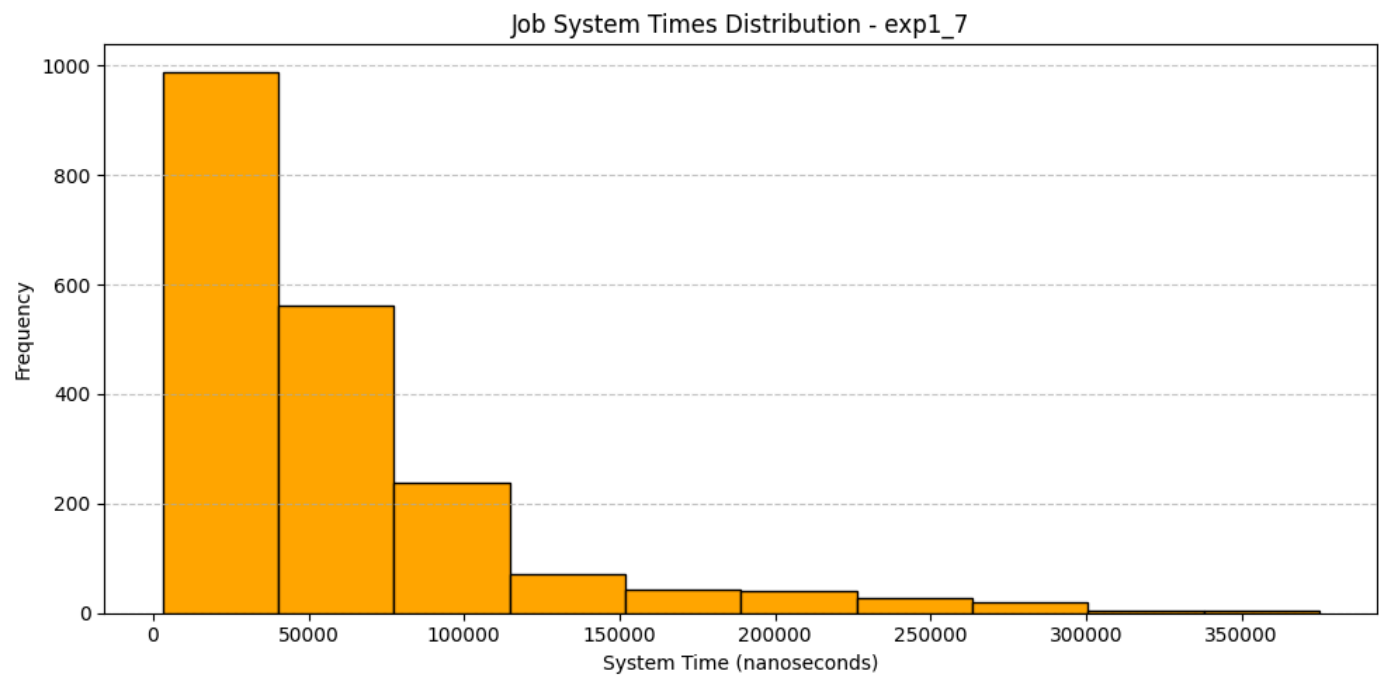


7. Single Client, Unbounded Queue ($\mu=50$, $\lambda=35$, Jobs=2000)

7. Queue Size Over Time

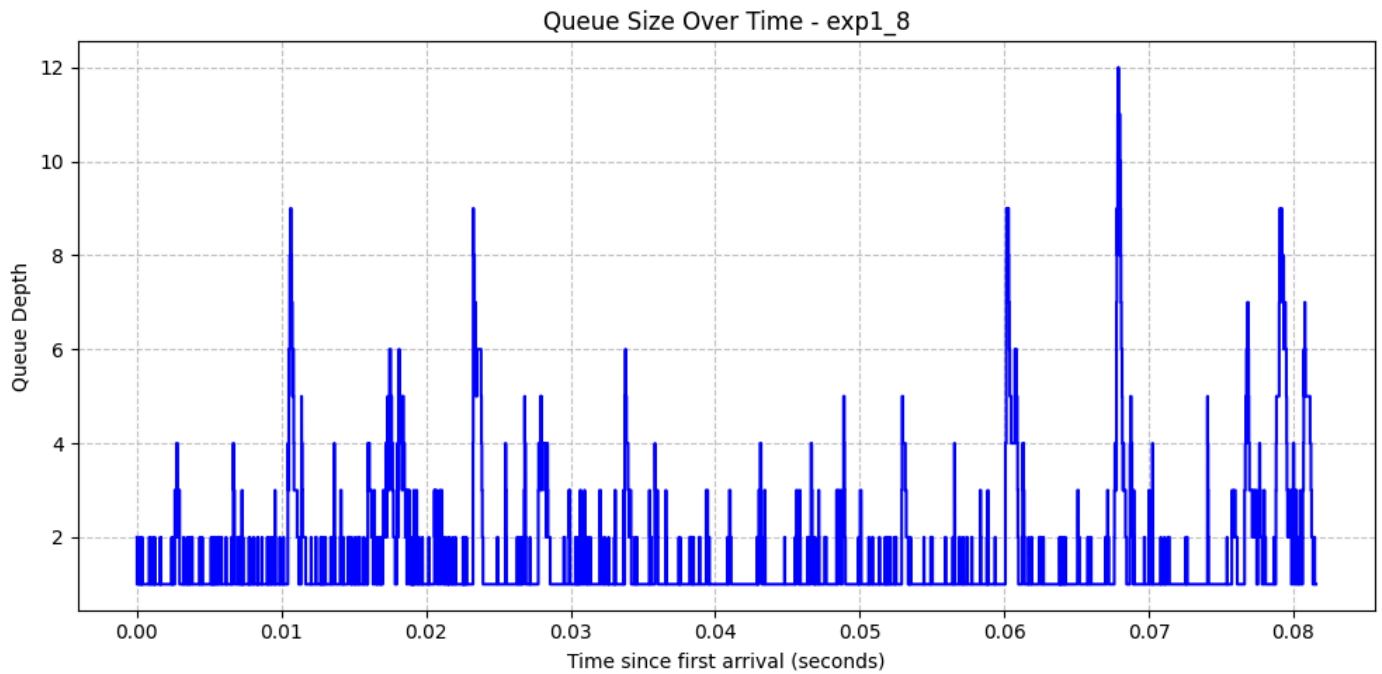


7. Job System Times Histogram

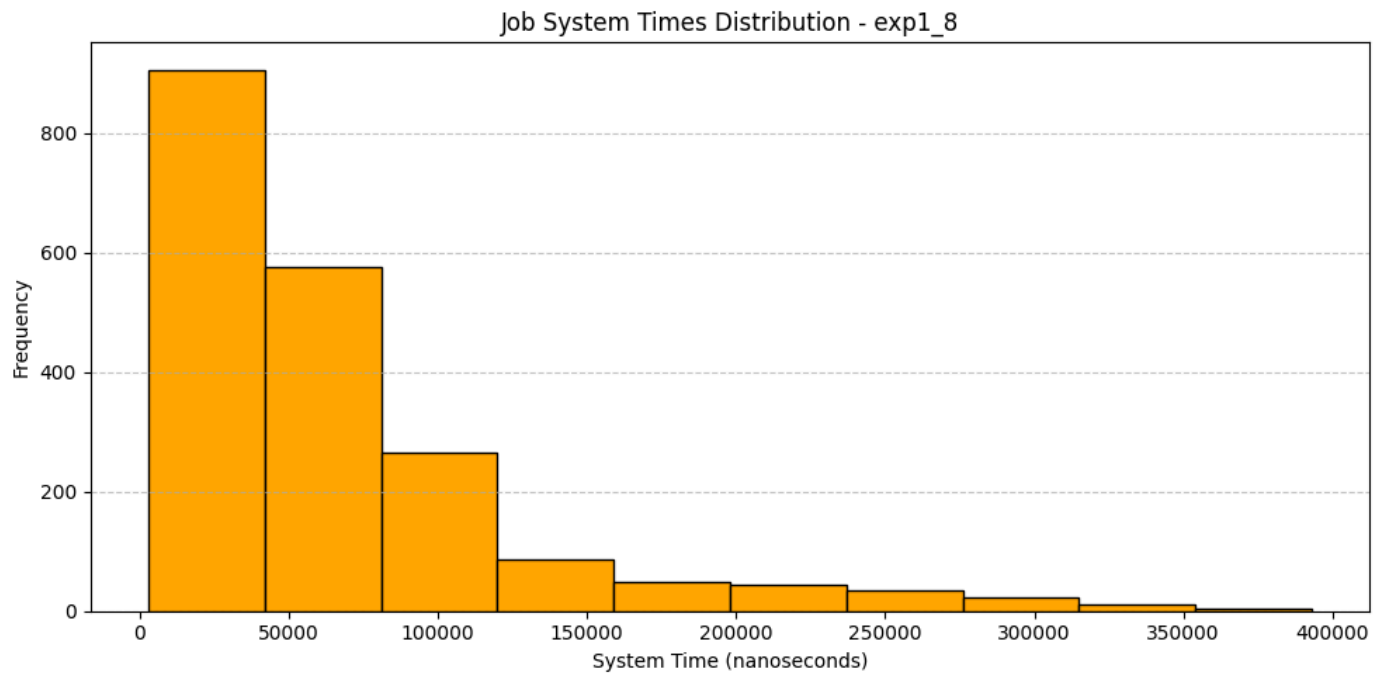


8. Single Client, Unbounded Queue ($\mu=50$, $\lambda=40$, Jobs=2000)

8. Queue Size Over Time

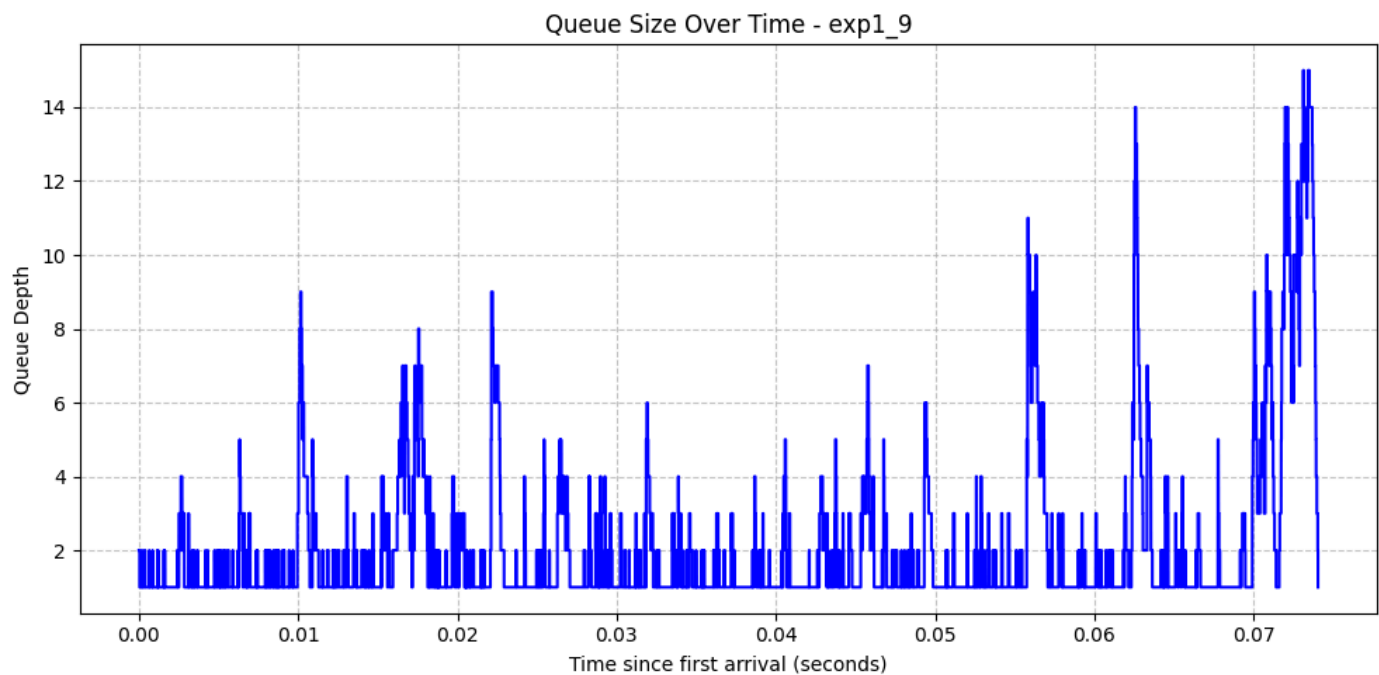


8. Job System Times Histogram

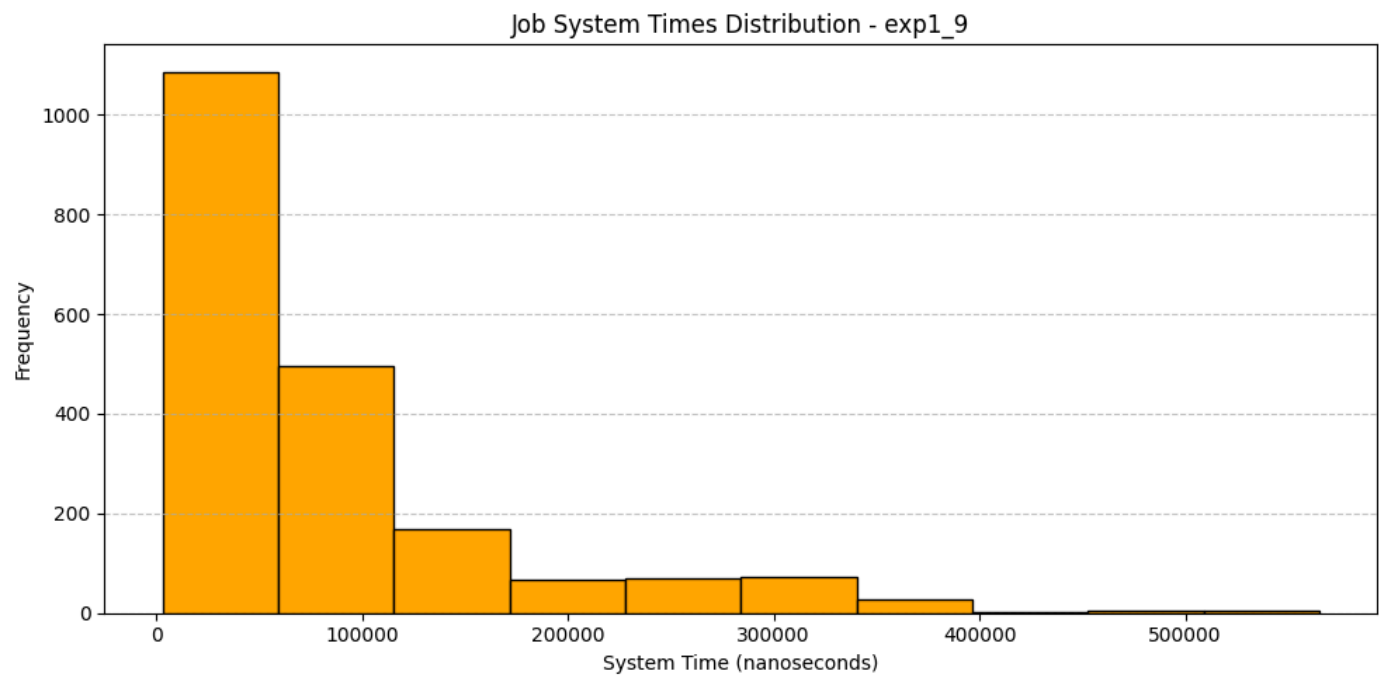


9. Single Client, Unbounded Queue ($\mu=50$, $\lambda=45$, Jobs=2000)

9. Queue Size Over Time

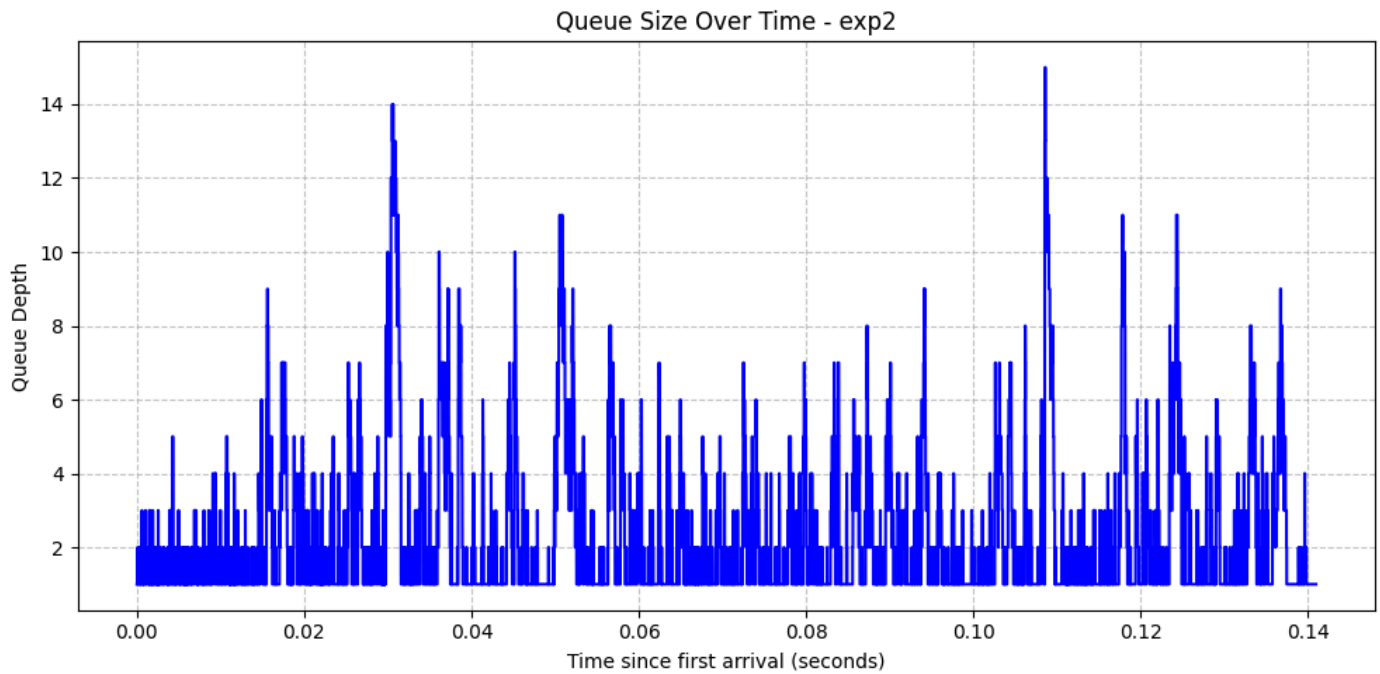


9. Job System Times Histogram

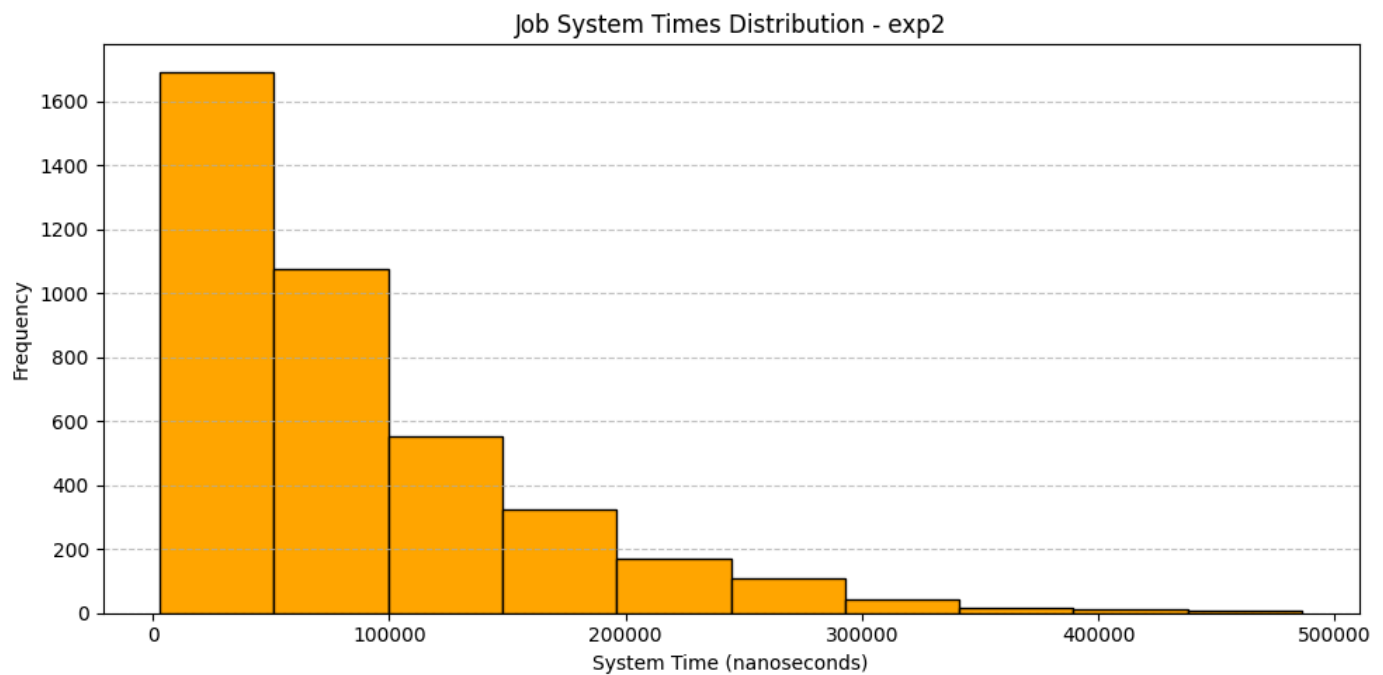


10. Two Clients, Unbounded Queue ($\mu=50$, $\lambda=20$, Jobs=4000 total)

10. Queue Size Over Time

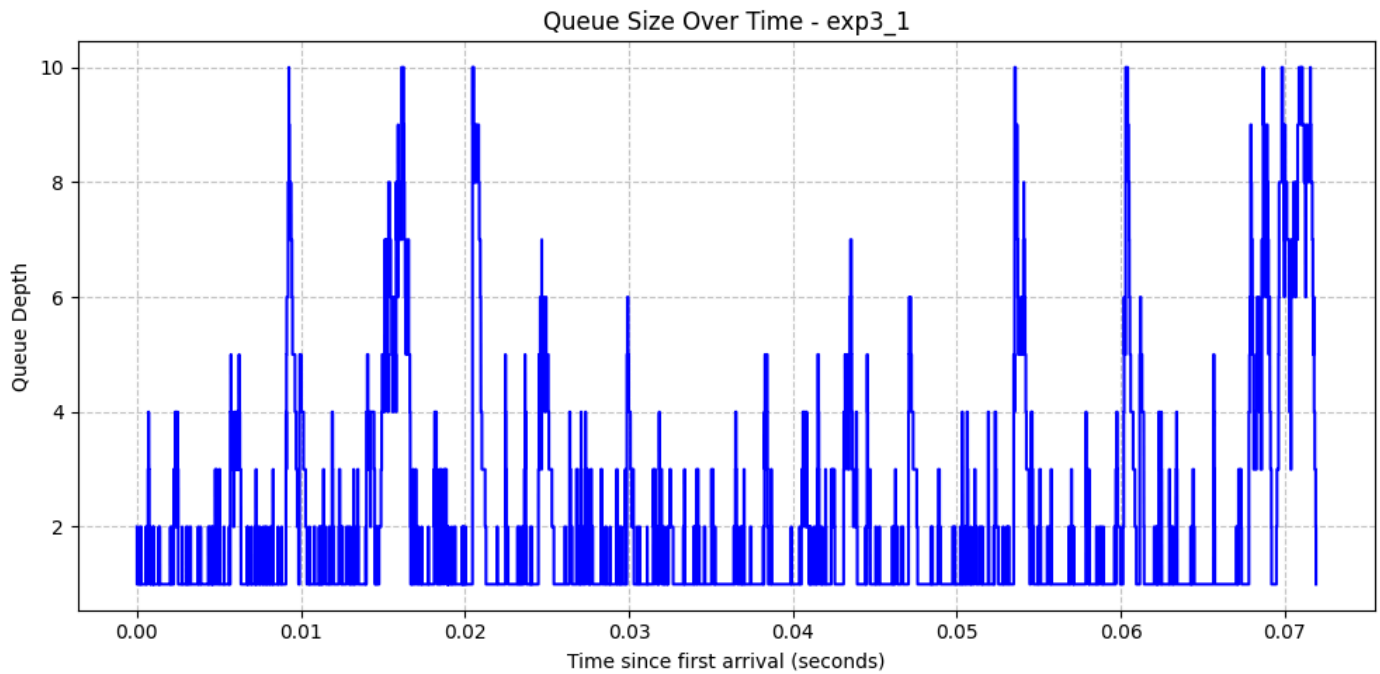


10. Job System Times Histogram

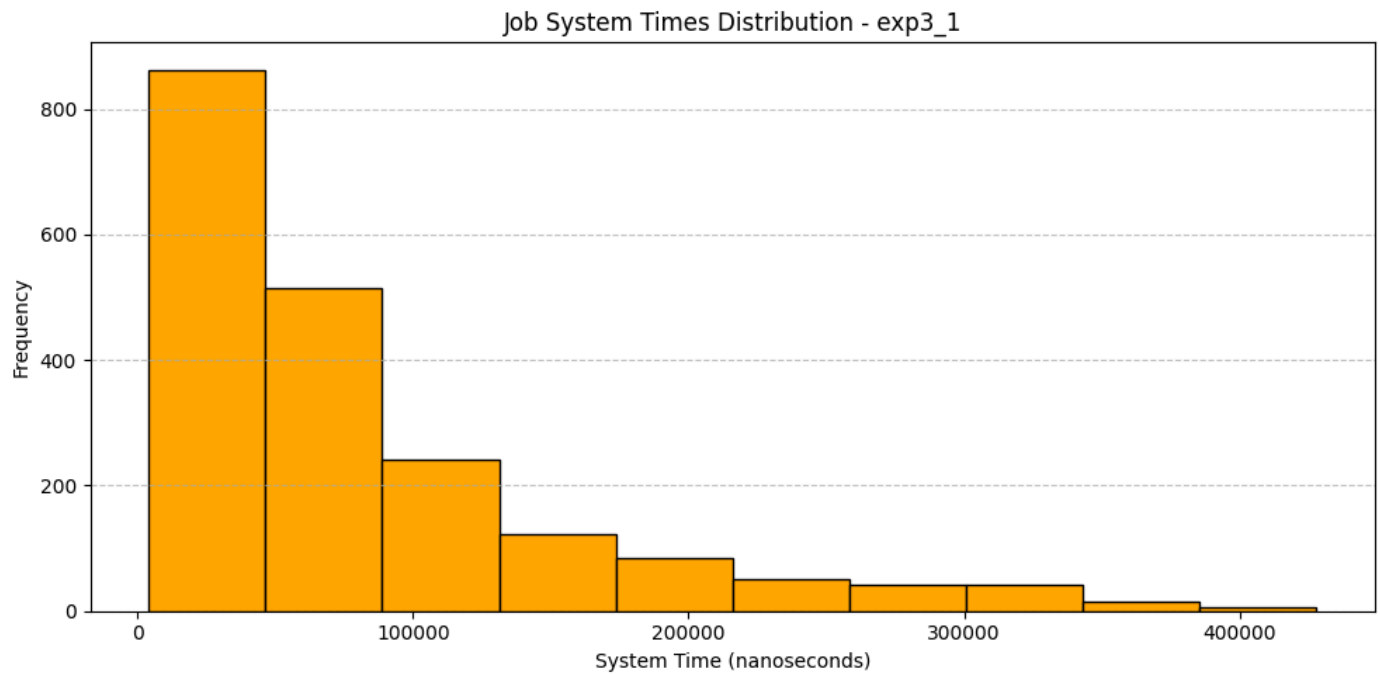


11. Single Client, Bounded Queue (Size=10, $\mu=50$, $\lambda=45$, Jobs=2000)

11. Queue Size Over Time

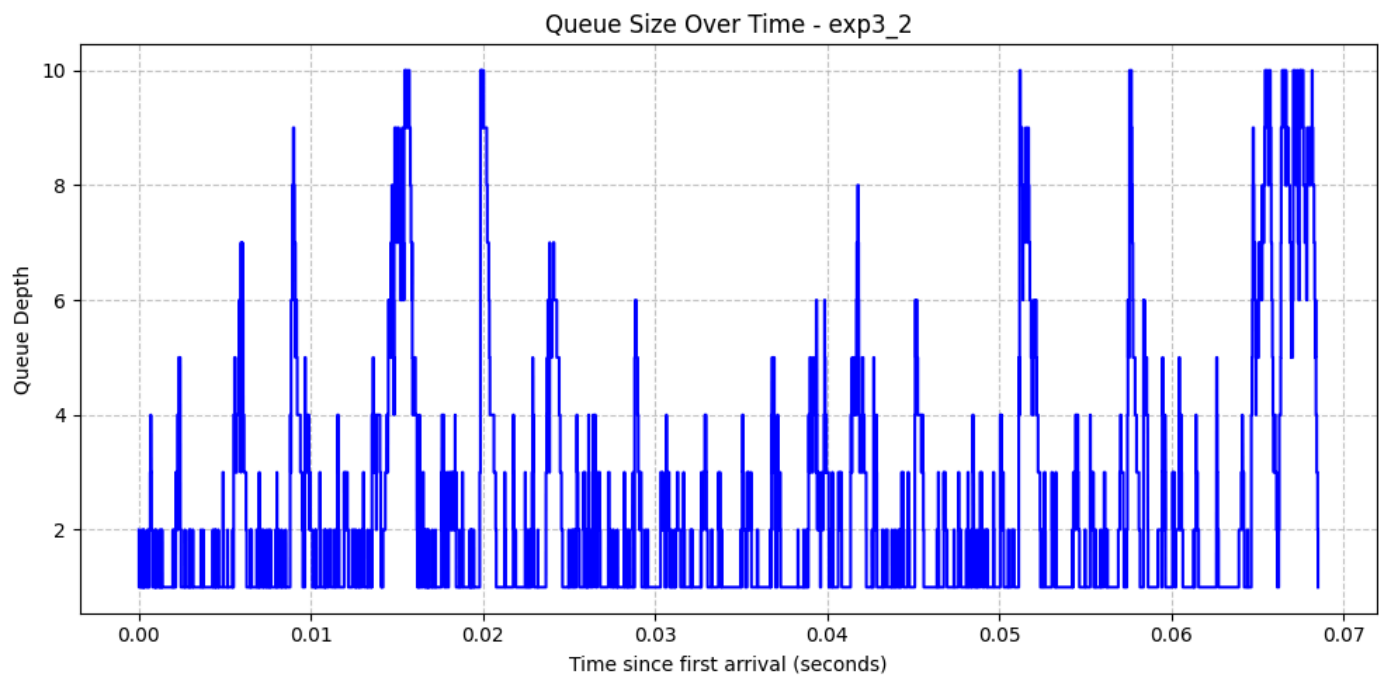


11. Job System Times Histogram



12. Single Client, Bounded Queue (Size=10, $\mu=50$, $\lambda=48$, Jobs=2000)

12. Queue Size Over Time



12. Job System Times Histogram

